AQUAFAB INDUSTRIES

Running a Temperature Program

Tidal cycles, ramps and recorded field traces.

For Aquafab Workbench 0.1.0

Edition of 28 October 2026

Screenshots show simulated machines and modelled example data, not measurements. Aquafab Workbench is not a safety system.

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1 Running a temperature program

A temperature program makes the chamber follow a curve: a tidal cycle, a ramp, a series of holds, or a temperature log recorded in the field. It can run on its own, recording temperature and flow, or alongside respirometry.

> Screenshots are from a simulated machine. Numbers in them are not measurements.

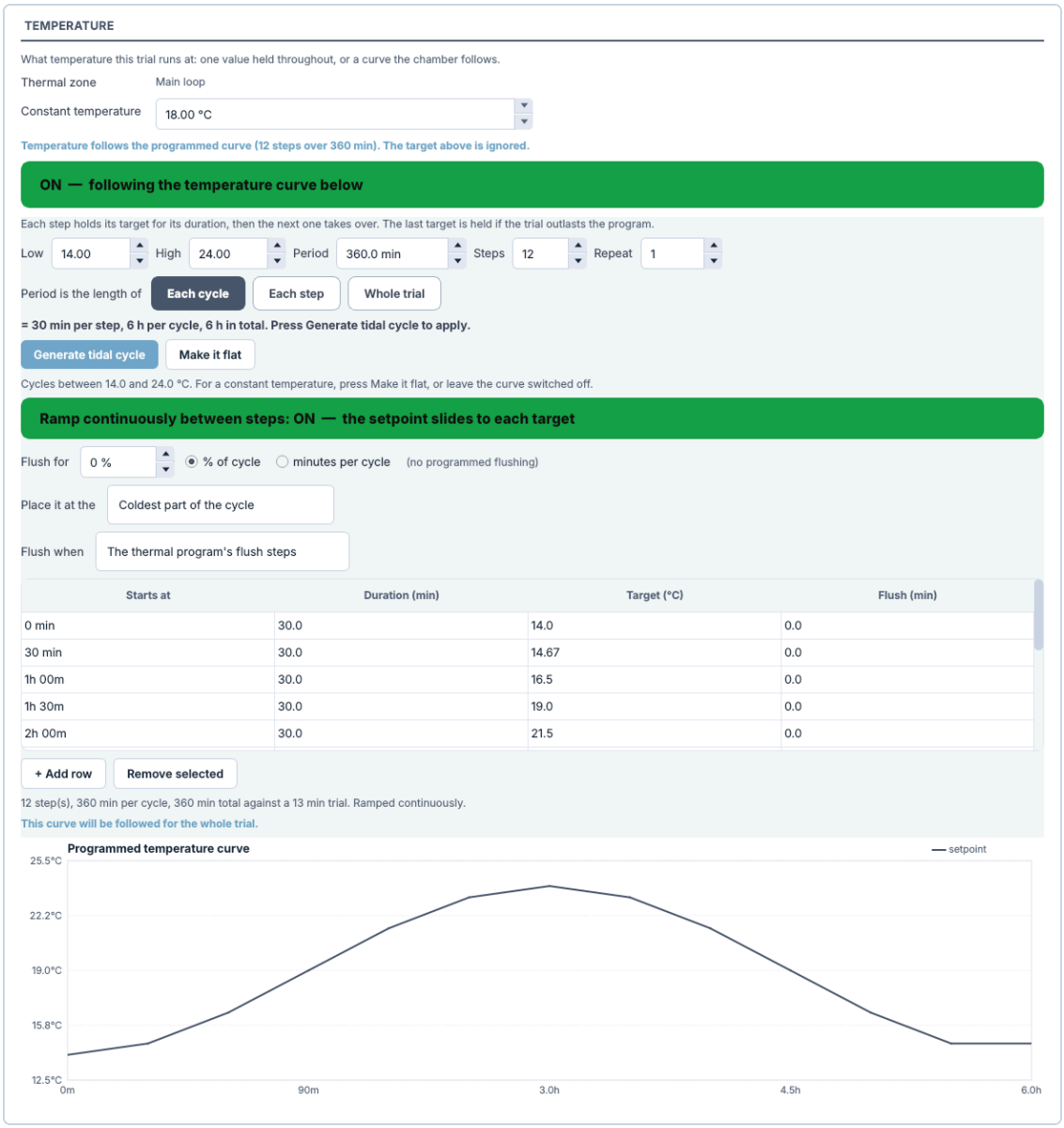
1.1 Before you start

Run the **Thermal Step Test** once on the machine (*Commissioning a New Machine*, step 4). It measures how fast the chamber can heat and cool, and the program page uses that to warn when a curve asks for more.

1.2 Build the curve

Open the machine, then **Experiments → Respirometry → Set up**.

1. In **What are you measuring?**, choose **Temperature program only**. (Or **Respirometry** to measure oxygen during the program.)
2. In **This experiment**, give it a **Name** and **Treatment**.
3. Go to the **Temperature** card and press the large **OFF — holding one temperature** button so it reads **ON — following the temperature curve below**.



The Temperature card with a tidal cycle generated.

1.2.1 Generate a tidal cycle

Field	What to enter
Low, **High**	The extremes of the cycle.
Period (min)	One full cycle, for example 720 for a 12-hour tide, unless you change **Period is the length of**.
Steps	How many holds the curve is cut into. More is smoother.
Repeat	How many cycles to run back to back.
Period is the length of	**Each cycle**, **Each step** or **Whole trial**. For an hour-long trial of 12 cycles: Repeat 12, **Whole trial**, Period 60. The line underneath shows the time per step and per cycle.

Press **Generate tidal cycle**. Leave **Ramp continuously between steps** ON for a smooth curve rather than a staircase; press it to OFF for a staircase. **Make it flat** gives a long hold at one temperature.

1.2.2 Edit by hand

The program table lists every step: **Starts at**, **Duration (min)**, **Target (°C)** and **Flush (min)**. Edit any cell; **+ Add row** and **Remove selected** change the steps. The preview chart redraws as you edit.

1.2.3 Flushing during a long program

A long program in a closed chamber drives the animal hypoxic. Set:

- **Flush for**: as **% of cycle** or **minutes per cycle**;
- **Place it at the**: *coldest part* (the tide returning), *warmest part*, *start*, *end*, or *spread evenly*;
- **Flush when**: the program's flush steps, when oxygen drops below a setpoint (needs a probe), or whichever comes first.

Flushing is suspended during respirometry measurement windows.

1.3 Check it against the machine

Under the table:

- **A rate warning**, "*This curve is steeper than measured... The water will lag behind it*", means a step asks for faster heating or cooling than the step test measured. Lengthen the period or narrow the range.
- **A reservoir note**, on machines with a chilled reservoir, says how cold to chill it before starting.

Set **Water speed** if the flow should be held or stepped through the program. Press **Save experiment** to keep it.

1.4 Run

1. Switch to **Live** and press **Prepare system**. It brings the chamber to the program's first temperature and chills the reservoir. Wait until ready.
2. Press **Start Trial**.
3. The **Programmed curve** chart shows the curve, the measured temperature over it, and how hard the heater, exchanger pump and chiller are working.
4. The run saves itself to Trial History after every cycle; there is nothing to press when it finishes.

1.5 Follow a recorded field trace

To make the chamber follow a temperature log from a tide pool or a logger:

1. **This experiment → Import readings from a file…** and choose the CSV. In the dialog, choose the **Time column** and its unit, tick the readings to import and say what each is measured in. The import is kept as a trial, marked *imported* everywhere. (Needs a license.)
2. In **Trial History**, **Inspect** the imported trial and press **Use as a temperature program…**.
3. Choose the **Temperature series**, how many **Steps** the trace becomes, and how many times to **Repeat** it. Press **Use as the temperature program**.
4. The curve appears on the Temperature card. Check it, then prepare and run as above.